Vol 16 Chapter 7 — Substrate-Symplectic Structure (Category 6, via Spin(5) ≅ Sp(2))

Lyra (Claude Opus 4.8); Elie joint (Bergman matrix-coefficient-sum support)

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Vol 16 Chapter 7: Substrate-Symplectic Structure (Category 6)

7.0 Position and the label

Ch 5 built the Bergman kernel algebra; Ch 6 the Casey #14 restriction sequence. Ch 7 adds a second, equivalent way to read the same K-types: as representations of a symplectic group. The substrate's internal symmetry SO(5) is, up to double cover, the rank-2 symplectic group — and that fact is not decorative. It gives every spinor K-type a symplectic 2-form and a Hamiltonian flow, which turns out to be the same object as the heat-semigroup time evolution from the Tier 0 framework.

Label pinned (Cal #242): "Cat 6" / "Category 6" denotes the sixth substrate-mechanism FORCING category in the Thursday 2026-06-04 taxonomy (the *symplectic* category), per `Lyra_Substrate_Symplectic_Crosslink_Spin5_Sp2_v0_1.md` Sec. 6. It does not refer to C_2 = 6. The five prior categories are: (1) K-type structural, (2) Lie-algebra, (3) Bergman/Shilov, (4) Mersenne/affine, (5) multipole. Category 6 is symplectic. This chapter is the SU candidate leg C26.

7.1 The accidental isomorphism Spin(5) ≅ Sp(2)

Standard Lie theory (Cartan classification): the rank-2 simple Lie algebras B_2 and C_2 coincide, so

$$\mathrm{Spin}(5) \;\cong\; \mathrm{Sp}(2) \quad (\text{compact form } \mathrm{USp}(4)), \qquad \dim = 10.$$

The check: dim SO(5) = 10 = dim Sp(2). ⍰ The 4-dimensional spinor of SO(5) is the 4-dimensional fundamental of Sp(2). This is the same $B_2 = C_2$ coincidence that fixes the substrate root system (B_2, $\rho = (5/2, 3/2)$) — here read on the group rather than the root side.

7.2 The substrate stabilizer is symplectic

The substrate isotropy group is K = SO(5) × SO(2). Passing to the double cover of the first factor:

$$\widetilde{K} = \mathrm{Spin}(5) \times \mathrm{SO}(2) \;\cong\; \mathrm{Sp}(2) \times \mathrm{SO}(2).$$

So the K-types V_(λ_1, λ_2) — which Ch 2 organized as SO(5)×SO(2) representations — are equally Sp(2)×SO(2) representations. The half-integer-λ_1 (spinor) K-types are genuine Sp(2) representations that are only projective for SO(5); the double cover is doing real work, not bookkeeping.

7.3 Spinor tower = symmetric powers of the symplectic fundamental

Identify the fundamental: V_(1/2,1/2) (dim 4) = the Sp(2) fundamental. The charged-lepton spinor tower is then the symmetric-power tower of that fundamental:

Generation	K-type	as Sp(2) rep	C_2	parity
gen-1	V_(1/2,1/2)	fundamental (Sym^1)	5/2	half-int (bulk)
gen-2	V_(3/2,1/2)	Sym^3	15/2	half-int (bulk)
gen-3	V_(5/2,1/2)	Sym^5	29/2	half-int (bulk)

This is the representation-theoretic anchor for Gate B of Ch 9: the Pochhammer length k in (5/2)_k is the Sym^k degree of the Sp(2) symmetric power, fixed by the generation index — *not* by N_c. The symplectic reading is what pins the Sym-degree and resolves the integer collision. (The gen-2 Sym^3 degree's "3" is the symmetric power, not color.)

The symplectic 2-form on the fundamental is $\omega = dq^1 \wedge dp^1 + dq^2 \wedge dp^2$, the canonical form Sp(2) preserves.

7.4 Two equivalent readings: spectral ↔ symplectic

Each K-type carries two operator structures that the accidental isomorphism identifies:

- Spectral (Categories 1–2): V_λ is a K-type with Casimir eigenvalue $C_2(\lambda)$; dynamics = Mehler/heat semigroup (Ch 5, Sec. 5.2).
- Symplectic (Category 6): V_λ carries ω and a Hamiltonian flow generated by H_B.

The identification is exact: the heat-semigroup evolution of the Tier 0 framework

$$\rho_{\mathrm{commit}}(\tau) = \exp(-\tau H_B / \hbar_{BST})$$

is the Hamiltonian flow of H_B = Casimir under the symplectic form ω, transported through Spin(5) ≅ Sp(2). Spectral time evolution and symplectic Hamiltonian flow are one mechanism in two languages. That equivalence — not any single number — is the chapter's content and the basis of the C26 universality claim: the symplectic structure exists on every K-type because Spin(5) ≅ Sp(2) is global, not K-type-specific.

7.5 Matrix-coefficient-sum backbone (Elie joint support)

The symplectic reading sits on the Bergman-as-matrix-coefficient-sum backbone (Elie, Ch 7 v0.2 support note; Ch 4). Each K-type's Bergman matrix coefficient has a Pochhammer norm (Elie Toy 3919):

K-type	dim	Pochhammer/Bergman norm
V_(1/2,1/2)	4	$3\pi/2$^g = $3\pi/128$
V_(3/2,1/2)	16	$21\pi/512$ = (g/rank^2)·$3\pi/2$^g
V_(5/2,1/2)	40	$567\pi/8192$

Under the symplectic reading these are the Sp(2) Sym^k invariant norms. The Sp(2) symmetric-power structure is what makes the norm tower a clean Pochhammer cascade (the (5/2)_k of Ch 5, Sec. 5.3). Elie's sum supplies the coefficients; the symplectic structure supplies the *reason they form a symmetric-power tower*.

7.6 Honest down-weight: the 1432/g coincidence (Cal #27)

The v0.2 source note floated a Gate-4 "resolution": a naive composition Pochhammer × Casimir × Weyl-ratio = 1432, divided by a symplectic factor 1/g = 1/7, gives 1432/7 = 204.6 ≈ 207 = m_μ/m_e.

This chapter explicitly down-weights that. It is 204.6 vs 206.768 = 1.05% — an order of magnitude worse than the Composite v0.5 form (0.112%, Ch 9 / F32). Two different mechanisms cannot both produce m_μ/m_e; the better-precision, better-forced Composite v0.5 form supersedes it. Moreover the "naive composition 1432" was never an independently motivated object, and dividing by g to land near 207 is exactly the pattern-match Cal #27 / Cal #35 warns against. The honest statement: the symplectic structure's value is structural (Sec. 7.3–7.4 — pinning Sym-degree, identifying flow with spectral evolution), not this numerical coincidence. The 1432/g reading is retired.

7.7 Cross-references

- Ch 2 (Lyra + Grace): K-type spectral decomposition — the Sp(2) symmetric-power reading pins the Sym-degree (Gate B)
- Ch 5 (Lyra): Pochhammer cascade (5/2)_k — Sec. 7.3 supplies its symmetric-power origin

- Ch 6 (Lyra): Casey #14 restriction — SO(4) ≅ SU(2)×SU(2) is the companion accidental isomorphism at codim-4
- Ch 9 (Lyra): Gate B (Sym-degree pin) is discharged here by the Sp(2) reading
- Ch 4 (Elie): matrix-coefficient backbone (Sec. 7.5)

7.8 Multi-week gates (Cal #189)

- Gate 7-1: explicit Spin(5) ≅ Sp(2) intertwiner on V_(1/2,1/2); show the SO(5) spinor action = Sp(2) fundamental action elementwise.
- Gate 7-2: prove the heat-semigroup = symplectic Hamiltonian flow identification (Sec. 7.4) at operator level, not just by parameter-matching.
- Gate 7-3 (C26 ratification): symplectic structure UNIVERSAL across all K-types — the global-isomorphism argument made rigorous. This is what promotes C26 from candidate to standalone SU leg.

7.9 Closure v0.1

Vol 16 Ch 7 scaffolded, Lyra primary. The chapter's claim: the substrate stabilizer K = SO(5)×SO(2) is symplectic (≅ Sp(2)×SO(2)), so every K-type carries a symplectic 2-form and a Hamiltonian flow, and that flow is the Tier 0 heat-semigroup evolution. The spinor tower is the Sp(2) symmetric-power tower — which pins the Pochhammer Sym-degree and discharges Ch 9 Gate B. "Category 6" = the symplectic substrate-mechanism category (pinned, not C_2=6). The 1432/g numerical coincidence for m_μ/m_e is explicitly retired (1.05% vs the superseding 0.112% Composite v0.5 form) per Cal #27.

Tier: Vol 16 Ch 7 scaffolding v0.1; Gates 7-1..7-3 multi-week per Cal #189; C26 candidate leg pending Gate 7-3.

— Lyra, Sat 2026-06-06 11:05 EDT. Vol 16 Ch 7 v0.1 Substrate-Symplectic (Category 6, Lyra primary): Spin(5) ≅ Sp(2) makes the substrate stabilizer symplectic; spinor tower = Sp(2) symmetric-power tower (pins Ch 9 Gate B Sym-degree, not N_c); heat-semigroup ρ_commit(τ) = symplectic Hamiltonian flow (two equivalent readings); "Category 6" pinned as the symplectic mechanism-category per source note, NOT C_2=6; 1432/g ≈ 207 coincidence RETIRED per Cal #27 (1.05% vs superseding 0.112% Composite v0.5); Elie matrix-coefficient-sum backbone folded in; Gates 7-1..7-3 + C26 ratification multi-week.